

J-ART: A Japanese Adversarial Red-Team Framework for Application-Layer LLM Security and Cost-Efficiency

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Abstract

Most open red-teaming and jailbreak benchmarks for large language models (LLMs) are English-centric and evaluate *bare models*. We present **J-ART** (Japanese Adversarial Red-Team framework), an open, reproducible harness that (i) evaluates the **application-structure layer** — model $\times$ system-prompt strength $\times$ guardrail $\times$ retrieval (RAG) — rather than bare models; (ii) applies **Japanese-specific obfuscation transforms** (gyaru-moji character substitution, vertical/newline splitting, keigo “polite-coercion” framing, double-talk false premises, Base64, and leet/zero-width smuggling) that defeat naïve keyword filters; (iii) grounds attacks in **MITRE ATLAS** (10 inference-time techniques); and (iv) reports a **cost-efficiency** metric alongside a defense rate with **Wilson 95% confidence intervals**. Attacks use *harmless proxy markers* (a fictitious canary and benign output markers); the framework contains **no weaponizable content**, and the malicious “core” of each attack is masked in all published artifacts. Across 21 application configurations and 11 attacks $\times$ 7 transforms (live, $K = 5$ trials per cell), we find that **application-layer defenses dominate raw model capability** — naked models range 29.6–86.5% defense, but a balanced ablation on **three** base models with **cluster-robust** intervals shows a **hardened prompt and model-based guardrails significantly raise defense on every model**, while lightweight keyword/regex filters help most where the base model is weak (effects measured against the naked baseline; the hardened arm saturates, so guardrail-atop-hardened is out of scope) — and that a simple **normalizing filter** defeats the obfuscation transforms a naïve keyword filter misses. We also document **large within-campaign variance**: in a controlled $K = 5$ repeat of a frozen configuration set, **9.6% of cells changed outcome across identical trials**, which makes single-sample leaderboards unreliable and motivates our statistical treatment.

Note. This is *not* an official safety evaluation of any vendor’s model; reported numbers are run-time snapshots subject to Section 7, and all defense rates are *lexical-leak* upper bounds (Section 3). The Section 4 numbers reproduce from committed result artifacts; for double-blind review, the code and artifacts are provided through an anonymized repository (Section 9) and will be released under open licenses upon acceptance.

1 Introduction

Public LLM red-teaming tooling (e.g., garak [2]) and jailbreak benchmarks (e.g., HarmBench [3], Jailbreak-Bench [4]) are predominantly English. Multilingual safety work has shown that *translating* prompts into low-resource languages can bypass safety training, but **language-specific obfuscation** — exploiting a language’s own scripts, typography, and register — remains comparatively underexplored. Japanese in particular offers a rich obfuscation surface (multiple scripts, full-width/zero-width characters, vertical writing, and conventionalized polite registers).

Two further gaps motivate J-ART. First, **bare-model versus deployed-system evaluation**: practitioners deploy a model *plus* a system prompt, *plus* guardrails, often *plus* retrieval; the security-relevant unit is the **application configuration**, not the model. Second, **security and cost jointly**: defenses cost tokens

or latency, so we report a cost-efficiency score so that “cheapest sufficiently-safe configuration” is directly readable.

Our contributions are: (a) a Japanese obfuscation transform suite; (b) an application-layer, MITRE-ATLAS-grounded harness with a safe proxy-marker methodology; (c) a guardrail comparison including a normalizing filter and a model-based classifier; (d) a statistically-reported leaderboard (Wilson CIs, optional repeated trials) with a cost-efficiency metric; and (e) an empirical observation of large router-induced run-to-run variance.

2 Related Work

J-ART sits at the intersection of six lines of work, summarized below and positioned in Table 1.

Adversarial red-teaming tooling. garak [2] scans a *model* with a library of vulnerability probes. J-ART instead treats the deployed *application configuration* (model $\times$ system prompt $\times$ guardrail $\times$ RAG) as the unit under test.

Jailbreak and harmful-behavior benchmarks. AdvBench/GCG [11], Jailbroken [12], HarmBench [3], JailbreakBench [4], and in-the-wild prompts [13] standardize *which* harmful behaviors to elicit and *how* to score them, predominantly in English at the model layer. J-ART reuses the harness idea but replaces harmful elicitation with a safe proxy-marker objective (§3) so the suite can be released publicly.

Automated attack generation. PAIR [6] and TAP [7] optimize jailbreak prompts against a target. J-ART uses a *fixed, auditable* transform suite, trading attack strength for reproducibility and safe disclosure.

Multilingual and low-resource safety. Deng et al. [14] and Yong et al. [5] show alignment degrades under *translation* into low-resource languages, and Wang et al. [15] study cross-lingual safety generalization. These study *translation across* languages; J-ART studies *intra-language orthographic obfuscation* specific to Japanese (gyaru-moji, vertical writing, keigo framing, kana/kanji encoding), which persists even though Japanese is high-resource.

Encoding/typographic obfuscation. Encoding-based jailbreaks are established prior art: Cipher-Chat [28] (ciphered conversation) and ArtPrompt [29] (ASCII-art typography). Our `base64_wrap` and `leet_smuggle` transforms are **language-agnostic replications** of this line, used only as a baseline; the novel surface is the Japanese-script transforms (§3).

Prompt injection and application-layer evaluation. Perez & Ribeiro [16] introduced goal-hijacking; Greshake et al. [8] formalized *indirect* injection; Liu et al. [17] benchmark LLM-integrated apps; Agent-Dojo [30] is the closest application-layer evaluation (tool-using agents). J-ART instead holds the *application configuration* (prompt $\times$ guardrail $\times$ RAG) as the explicit unit of evaluation against a Japanese-obfuscation surface.

Guardrails and input classifiers. Llama Guard [9], NeMo Guardrails [18], and the OpenAI Moderation API [19] provide model- or rule-based filtering. J-ART treats the guardrail as a *swappable configuration variable* and measures the marginal defense each adds, including a deterministic normalizing filter that defeats obfuscations a naive keyword filter misses (§4).

LLM-as-judge validation. Zheng et al. [20] and the survey of Gu et al. [21] document the biases of LLM judges. J-ART avoids an LLM judge for its *primary* outcome, using a deterministic canary/marker check (§3) so judge variance does not enter the headline metric.

Japanese NLP resources and taxonomies. General Japanese benchmarks such as JGLUE [22] cover NLU, but to our knowledge no open, reproducible *adversarial-safety* benchmark targets Japanese-specific

Table 1: Positioning of J-ART relative to representative prior work.

Work	Language	Unit of eval.	Cost axis	Guard/RAG	Public
garak [2]	English	Model	No	No	Yes
HarmBench / JBB [3; 4]	English	Model	No	No	Yes
PAIR / TAP [6; 7]	English	Model	No	No	Partial
Multiling. [5; 14; 15]	Low-resource	Model	No	No	Yes
Llama Guard / NeMo [9; 18]	Eng.-centric	Classifier	No	n/a	Yes
J-ART	Japanese	App config	Yes	Yes	Yes

Table 2: Classification of obfuscation transforms.

Class	Transforms	Exploits
Japanese-specific	gyaru , vertical_newline , polite_business , double_tongue	Kana glyph variants, vertical writing, keigo authority/in-group framing
Language-agnostic	base64_wrap , leet_smuggle (+ baseline)	Encoding / symbol substitution applicable to any language

obfuscation — the gap J-ART addresses. We map attacks to MITRE ATLAS [1] and cross-reference the OWASP Top 10 for LLM Applications [10].

J-ART is thus distinguished by combining a *Japanese intra-language obfuscation* surface, an *application-configuration* unit of evaluation, a *cost-efficiency* axis, and a *safe proxy-marker* methodology in a single reproducible harness.

3 Methodology

Application-structure layer. Each evaluated **target** is a configuration: provider/model $\times$ prompt_strength $\in$ {naked (low), hardened (high)} $\times$ guardrail $\in$ {none, keyword, regex, llm, llamaguard} $\times$ rag $\in$ {true, false}. A simplified RAG context (benign corporate documents, optionally poisoned for indirect-injection attacks) and a system prompt containing a secret *canary* are assembled per trial.

Japanese obfuscation transforms. Each base attack is rendered through one of seven transforms: **baseline** (none), **polite_business** (keigo authority framing), **vertical_newline** (one-character-per-line vertical writing), **gyaru** (gyaru-moji glyph substitution + zero-width spaces), **double_tongue** (false sandbox/authorization premise), **base64_wrap** (Base64-encoded instruction), and **leet_smuggle** (leet symbol substitution + zero-width spaces). Transforms preserve human legibility while breaking substring-matching filters.

We distinguish two classes (Table 2). *Japanese-specific* transforms exploit the Japanese writing system or register and have no direct English equivalent: **gyaru** (kana glyph substitution), **vertical_newline** (vertical writing), and **polite_business/double_tongue** (keigo authority and in-group framing). *Language-agnostic* transforms — **base64_wrap** and **leet_smuggle**, plus the **baseline** control — are **not novel**: they replicate the encoding/typographic jailbreak line (CipherChat [28], ArtPrompt [29]) and serve only as a baseline so the marginal effect of the Japanese-specific surface can be isolated.

Attack suite and MITRE ATLAS mapping. Eleven attacks cover ten inference-time ATLAS techniques: Direct/Indirect Prompt Injection (AML.T0051.000/.001), LLM Jailbreak (AML.T0054), LLM Data Leakage (AML.T0057), LLM Prompt Self-Replication (AML.T0061), Trusted-Output/Citations

manipulation (AML.T0067.000), LLM Prompt Obfuscation (AML.T0068), System-Prompt discovery (AML.T0069.002), RAG Poisoning (AML.T0070), and False RAG Entry Injection (AML.T0071). Vectors are **user** (direct) or **rag** (indirect).

Harmless proxy markers and safety-by-construction. To avoid producing weaponizable content, attacks target **harmless proxies**: a fictitious canary passphrase embedded in the system prompt (data-leakage attacks) and benign string markers [JART-PWNED] / [JART-HIJACKED] (injection / control-hijack attacks). The malicious instruction **core** of each attack is defined in code, kept out of configuration files, and **masked** (replaced by a redaction placeholder) in **results.json** and the public site; only the obfuscation *framework* is shown, so developers can study the structure without a usable recipe. A defense-in-depth client-side mask is also applied at render time.

Deterministic judge. A breach is declared by membership — the canary or marker in the response, matched exactly *and* after light normalization. This judge detects a *lexical* leak and cannot detect a semantic/paraphrased one, so we call the metric a **lexical-leak defense rate**: every defense rate in this paper is an **upper bound** on true safety, not a certificate. A human-validated judge is future work.

Guardrails. Five settings are compared: **none**; **keyword** (banned-substring match — fast, weak to obfuscation); **regex** (a *normalizing* filter that strips zero-width characters, reverses leet substitutions, folds whitespace, and Base64-decodes long tokens *before* keyword matching — deterministic, zero extra API cost); **llm** (the target model self-moderates input and screens output); and **llamaguard** (a dedicated safety-classification model — Llama Guard 4 via a model router — classifies the input as attack/benign, with a graceful fallback on failure).

Statistical treatment. Each defense rate carries a **Wilson 95% confidence interval** over its trial count; each cell (target, attack, transform) is repeated N times (JART_TRIALS). The trial-level Wilson interval treats the K repeats of a cell as independent, which they are **not**. For any *difference* between configurations (the ablation, §4) we therefore report a **cluster-robust bootstrap 95% CI** that resamples whole *cells* (**bootstrap_diff_ci**, 3,000 resamples, fixed seed); these are wider than the trial-level Newcombe intervals and are the intervals on which every significance claim rests.

Cost-efficiency metric. We report $\text{cospa} = \text{defense_rate}(\%) \div \text{cost_per_million_tokens (USD)}$ (per-million cost clamped at a \$0.01 floor). *cospa* is a deliberately simple *screening ratio*, not a utility model: its units are not independently meaningful and the ranking is sensitive to the clamp and price snapshot. We treat it only as a sort key and present the underlying **defense-vs-cost trade-off** directly; the clamp affects only sub-\$0.01/Mtok configurations, of which our fleet has none.

Execution. Trials are independent and run with bounded concurrency (a thread pool, default 8 workers) under per-request timeouts and exponential-backoff retries; determinism of the simulation does not depend on execution order. Targets whose API key is absent fall back to a deterministic simulation so a partial key set never aborts the run.

4 Results

We report a fully-labeled **live campaign** (21 configurations $\times$ 11 attacks $\times$ 7 transforms $\times$ $K = 5$ trials = 8,085 trials), executed 2026-06-14. Every cell is labeled **mode=LIVE** with a price snapshot; a trial is excluded iff **api_error** (at the trial level); after parity re-runs this removes 169 of 8,085, leaving 7,916 **clean trials**; billed cost over clean cells (campaign plus parity re-runs, including the expensive Opus flagship) is $\approx$ \$6.0. **Opus naked is now fully measured** ($n=385$, after an Anthropic top-up); **only Gemini-pro remains partial** ($n=227$; per-minute rate limits) and is treated as *provisional*. All defense rates are **lexical-leak** upper bounds (§3); every difference carries a **cluster-robust CI** (§3). Each §4 number is reproducible from committed artifacts (§9).

Table 3: Naked-model defense (no prompt, no guardrail; $K = 5$).

Naked model	Defense	95% CI	n
DeepSeek-V3	29.6%	[25.3, 34.4]	385
Llama 4 Scout	37.9%	[33.2, 42.9]	385
Gemini 2.5 Flash	49.6%	[44.6, 54.6]	385
gpt-oss-120b	82.6%	[78.5, 86.1]	385
GPT-4o-mini	83.9%	[79.9, 87.2]	385
Claude Opus 4.x	86.5%	[82.7, 89.5]	385

Table 4: Configuration effect (Δ vs. that model’s naked baseline; cluster-robust 95% CI). $\times$ = not significant.

Added defense (low-prompt)	Scout (38.1%)	4o-mini (82.3%)	GPT-4.1 (64.1%)
+ keyword	+26.0 [+13,+39]	+9.1 [−1,+20] $\times$	+12.5 [−1.5,+26] $\times$
+ regex	+45.0 [+33,+56]	+10.0 [+0,+20] $\times$	+23.8 [+11,+36]
+ Llama Guard	+47.2 [+36,+58]	+15.6 [+7,+25]	+25.1 [+13,+38]
+ LLM guardrail	+59.7 [+51,+68]	+17.7 [+10,+26]	+35.9 [+26,+47]
hardened prompt	+47.2 [+36,+58]	+17.7 [+10,+26]	+33.3 [+23,+45]

Overall. All configurations include the simplified RAG context (§3); **“naked” means no system-prompt hardening and no guardrail, not “no RAG”** (indirect-injection attacks are in scope even for the naked baselines). The aggregate breach rate over clean trials was **14.5%** (1,151/7,916), **concentrated in weak configurations**. (gemini-pro contributes a *provisional* $n=227$ of defended trials to this pool, §7; excluding it raises the aggregate to 15.0%, leaving the picture unchanged.) Every hardened or guardrailed configuration, on *any* model, defended in the **91–100%** band, whereas bare models ranged from **29.6% to 86.5%** (Table 3).

Application-layer defense vs. model capability. We isolate the configuration effect from model capability with a balanced ablation on **three** base models spanning the range (weak Llama-4-Scout; cheap GPT-4o-mini; flagship GPT-4.1). Each holds the model fixed and crosses {naked, hardened} prompt $\times$ guardrail; we report each config’s difference from that model’s naked baseline with a **cluster-robust bootstrap 95% CI** (3,000 cell-resamples; resampling cells, not trials, so the within-cell correlation of the ablation’s $K=3$ repeats does not inflate significance; reproduced by `experiments/analyze_ablation.py`; Table 4). Two robust patterns hold across all three models: (1) a **hardened prompt** and the **model-based guardrails** (Llama Guard, LLM self-moderation) significantly raise defense on *every* model; (2) the effect **scales inversely with the naked baseline** — on weak Scout (naked 38%) every guardrail adds +26 to +60 points, whereas on GPT-4o-mini (naked 82%, near the ceiling) the lightweight keyword/regex filters are *not* individually significant under clustering. Configuration matters most exactly where the base model is weakest. The thesis — *configuration matters more than model choice* — rests on a clean, fully-measured comparison: a hardened cheap model (GPT-4o-mini hardened $\approx$ 100%) **out-defends a naked flagship** (GPT-4.1 naked 64.1%, $n=231$) by ~ 36 points, and likewise out-defends a naked Opus (86.5%, $n=385$). Even at full parity the strongest naked model trails a hardened cheap one.

Within-campaign trial variance (a primary result). Within a single campaign, every cell of a frozen configuration set is sampled $K = 5$ times. Over the **1,577 cells with all five trials clean, 9.6%** (152/1,577) **changed outcome across the five identical repeats**, concentrated in naked weak models. For “Llama 4 Scout / naked” — the original cautionary example — of its 77 cells, **32 (42%) were unstable**: 14 never breached (0/5), 31 always (5/5), and 32 in between. A *single-sample* leaderboard would assign this configuration anywhere from 0% to 100% by chance — controlled evidence (free of the configuration-set confound) that **a single sample per cell is not reproducible**. We scope this to *within-campaign* (provider non-determinism); across-day drift is scaffolded (`across_runs`) but left to future work.

Guardrail comparison and cost. Across the three ablation models (Table 4) the ordering is consistent: the LLM guardrail and a hardened prompt give the largest gains, Llama Guard next, and lightweight keyword/regex help most on weak base models. Guardrails that block an input **skip the main model call**, lowering net cost while raising defense; the **normalizing regex filter** (verified in unit tests) adds +45 points on weak Scout at zero API cost. On the cost-efficiency metric, leaders are inexpensive OSS models behind a lightweight guardrail (gpt-oss-20b + keyword, cospa $\approx$ 1131), while a flagship naked configuration is worst (Opus naked, cospa $\approx$ 6.8) — three orders of magnitude apart.

Attack and transform effectiveness. The most effective attacks were Trusted-Output/Citations manipulation, False RAG Entry Injection, and System-Prompt discovery. Among transforms, the *un-obfuscated baseline* was most effective (**22.0%**), then *vertical_newline* (20.0%) and *polite_business* (17.3%), while **base64_wrap was least effective (5.5%)**: heavy encoding makes models refuse or fail to parse the instruction more often than it smuggles it past them. The most dangerous Japanese transforms are the *legible* ones (vertical writing, keigo framing), not the cryptographic ones.

5 The Leaderboard Artifact

The harness emits `results.json` and a self-contained static site (sortable leaderboard, per-configuration attack logs with masked prompts, JP/EN UI with environment-based auto-selection, per-model defense-rate CIs, and per-cell breach rates), continuously deployed.

6 Ethics and Responsible Disclosure

Safety by construction. J-ART is safe by construction. (1) Attacks target a **fictitious canary** and **benign markers**, never real harmful capabilities; no weaponizable content exists in the code, data, or published site. (2) The malicious instruction **core** is masked in all artifacts; only the obfuscation framework is shown. (3) Models are referenced for research comparison only; results are run-time snapshots, not certifications. The intended use is to help developers harden their *own* deployments.

A reusable disclosure pattern. We state the design above as a transferable pattern for publishing adversarial safety benchmarks without releasing operational attacks: (i) a *harmless proxy objective* (a fictitious canary and benign markers) so a breach is measurable but inert; (ii) a *core/framework split with masking* — publish the obfuscation framework that carries the research signal, mask the operational core in every artifact, and keep harmless cores only in code; (iii) a *deterministic offline mode* so results reproduce from the repository without distributing a working exploit corpus; and (iv) *audited masking* — a unit-test invariant checks that no core (raw or obfuscated) reaches a published artifact. This keeps the methodology and aggregate findings fully open while the attack payloads stay withheld — the posture we recommend for language-specific red-team datasets generally.

Governance crosswalk. Table 5 cross-references each evaluated capability to MITRE ATLAS [1], the OWASP Top 10 for LLM Applications [10], the NIST AI RMF [23] and its Generative-AI Profile [24], the EU AI Act [25], and Japan’s AI governance guidance [26; 27]. The mapping is **informational, not a compliance claim**; applicability depends on each deployment’s risk classification. Operationally J-ART is a **MEASURE**-stage activity in the NIST AI RMF whose outputs inform an organization’s **MANAGE** decisions.

Worked example. Under NIST MEASURE 2.7, an organization could record a J-ART line item as “prompt-injection resilience (T0051/T0054), lexical-leak defense rate 93.5% (Wilson 95% CI [90.6, 95.6] for this single rate; the cluster-robust interval is for *differences*, §3), $n=385$, $K=5$, GPT-4.1-mini + hardened prompt + keyword guardrail, 2026-06-14” with a pre-registered threshold (e.g. “CI lower bound $\geq$ 90%”); the same row is the evidence an AISI red-team report or an EU AI Act Art. 15 dossier would cite.

Dual-use. The published transform suite is also a reusable attack wrapper. The attacker uplift is low (the Japanese-script transforms are folklore; the encodings replicate public work; no harmful payloads are in-

Table 5: Governance crosswalk (informational).

J-ART (ATLAS)	capability	OWASP LLM Top 10	NIST AI RMF	EU AI Act	Japan
Prompt injection / jailbreak (T0051, T0054)		LLM01	MEASURE 2.7	Art. 15, 55	AISI; AI Biz
Canary / data leakage (T0057)		LLM02	MEASURE 2.7	Art. 15, 10	AI Biz
System-prompt discovery (T0069.002)		LLM07	MEASURE 2.7	Art. 15	AISI
RAG poisoning / false entry (T0070, T0071)		LLM08; LLM01	MEASURE 2.7	Art. 15, 10	AISI
Trusted-output manipulation (T0067.000)		LLM09; LLM05	MEASURE 2.6	Art. 50	AI Biz
Obfuscation / self-replication (T0068, T0061)		LLM01	MEASURE 2.7	Art. 55	AISI

cluded), while the defender benefit is concrete: the same repository ships the **normalizing regex guardrail** that neutralizes every transform (unit-tested). We judge open release net-positive for defenders, consistent with §6.2.

7 Limitations

- **Proxy markers, not harm:** we measure instruction-violation / leakage proxies rather than real-world harm.
- **Deterministic judge:** the judge matches the canary/marker exactly *and* after light normalization (case-folding, whitespace/zero-width removal), so a reformatted echo is counted as a breach. It cannot detect *semantic*/paraphrased leakage, so reported defense rates are an **upper bound**; a human-validated LLM-judge is future work.
- **Single-turn:** multi-turn / crescendo attacks are not yet modeled.
- **Sample size:** a handcrafted suite (11 attacks $\times$ 7 transforms); default $N=1$ per cell.
- **Provider non-determinism / router variance:** results are snapshots and depend on routing and model versions at run time.
- **Non-random missingness (now mostly resolved):** we flagged that an earlier campaign’s partial configs had **non-random** missingness — rate limits correlate with attack difficulty, under-representing Opus’s hardest attacks. Parity re-runs **confirmed this directly**: Opus naked at full $n=385$ defends **86.5%**, vs. the survivorship-biased **89.6%** on the $n=125$ earlier partial snapshot (not retained as an artifact; only the corrected $n=385$ figure is committed) — a ~ 3 -point over-estimate, as predicted. Qwen3-max and Opus are now at parity; **only Gemini-pro remains partial** ($n=227$; per-minute rate limits a credit top-up cannot raise) and is treated as provisional, with no headline resting on it.
- **Cost attribution:** guardrail tokens are now priced at the *guardrail model’s* own rate (the classifier JART_GUARD_MODEL for llamaguard; the target model itself for the llm guardrail, which genuinely calls it) and reported separately. Remaining approximation: simulated token *counts* are estimated.
- **Simulation mode** is a deterministic stand-in for missing keys and must not be read as empirical model behavior.

8 Future Work

LLM-judge with human-validated agreement; multi-turn and adaptive (PAIR/TAP-style) attacks; larger and partially human-curated attack sets; additional ATLAS techniques (cost-exhaustion, model extraction);

per-provider concurrency and incremental caching with longitudinal trend tracking; and dropping simulation mode for a purely empirical, repeated-trial statistical report.

9 Reproducibility

Code, configuration, the harness, and the committed result artifacts are provided to reviewers through an **anonymized repository** and will be released under open licenses (code MIT; report text CC-BY-4.0) upon acceptance. The harness is deterministic in simulation mode; live results depend on provider routing and model versions at run time and are reported with the run date and model identifiers. Each cell carries provenance labels — `mode` (LIVE/MOCK), `api_error`, the resolved model, and a `price_per_million` snapshot — and simulated cells are excluded from the live aggregates. **Every \$4 number is reproducible from committed artifacts:** `var_run.json` (§4.1 overall, naked-spread, variance, transforms; cospa); and the three ablation files `ablation_results.json` (GPT-4o-mini), `abl_scout_results.json` (Llama-4-Scout), `abl_gpt41_results.json` (GPT-4.1) for Table 4. Aggregates and cluster-robust CIs are recomputed by the provided `analyze_ablation.py` and `analyze_variance.py`. A continuously-deployed public leaderboard exists; its URL is withheld for double-blind review and provided in the camera-ready.

AI Writing Assistance

The author used **Claude Opus 4 (Anthropic)**, accessed via the Claude Code command-line interface, for writing and editing assistance during the preparation of this manuscript: (1) drafting and revising prose across three rounds of reviewer-driven revision; (2) bilingual editing of the Japanese-language abstract and the parallel Japanese version (`technical-report.ja.tex`); and (3) citation error checking (author attributions, publication years, bibliography completeness). All AI-generated text was reviewed, corrected where necessary, and approved by the sole author. The scientific content — experimental design, all live API evaluations, data collection, statistical computation (Wilson and cluster-robust intervals), and interpretation — was conducted independently by the human author without AI generation assistance. *Dual-role note:* Claude Opus 4.x also appears in this paper as one of the *evaluated* models (§4, naked defense rate 86.5%, $n=385$); its role as a writing tool and its role as a research subject are distinct and both disclosed here. The author takes sole and full responsibility for all content.

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